Zheng Jiang

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Education

2023/10-2027/06 **University of Cambridge**

BA and MMath Mathematical Tripos. Part II ranking: 20/240.

Relevant Courses - Algebraic Topology, Algebraic Geometry, Differential Geometry, Complex Geometry, Galois Theory, Representation Theory, Functional Analysis, Probability and Measure, Advanced Probability, General Relativity, Methods, Classical Dynamics, Fluid Dynamics, Statistics, Introduction to Deep Learning.

Research and Expository Work

2026/06-Present **General Relativity Research** - supervised by Prof. Mihalis Dafermos

- Studied O'Neill's *Semi-Riemannian Geometry with Applications to Relativity,* Hawking and Ellis's *The Large Scale Structure of Space-Time,* and Penrose's *Gravitational Collapse and Space-Time Singularities.* Read Carter's *Black Hole Equilibrium States,* Mazur's *Black Hole Uniqueness Theorems,* and modern treatments including *Stationary Black Holes: Uniqueness and Beyond.*
- Wrote two expository essays. One expanded proof details for the Penrose singularity theorem in the Part III Black Holes notes and explained the geometric intuition. The other traced Carter and Robinson's classical no-hair results through vacuum Kerr uniqueness to Mazur's electrovacuum proof of Kerr-Newman uniqueness, which rules out disjoint solution families under its hypotheses.

2025/08-2025/10 **Topology Research** - supervised by Dr. Gareth Wilkes

- Studied Lee's *Introduction to Smooth Manifolds,* Chern's *Lectures on Differential Geometry,* Hatcher's *The Kirby Torus Trick for Surfaces,* and Milnor's *Morse Theory.*
- Wrote an expository essay on the uniqueness of smooth structures on the real line and plane, explaining the role of Morse functions and handle decompositions in the proof for the plane.

Awards and Honours

2024/11-Present **Scholar at Selwyn College; Tripos Prize and Ratcliffe Prize** - awarded for First Class results in Part IA and Part II Tripos.

2026/10 **QRT MathQ** - ranked 3/52 among all teams in Cambridge.

2024/12 **ICMC (Imperial Cambridge Mathematics Competition)** - qualified for the final round.

2024/06 **Alibaba Global Mathematics Competition** - 274/50,000+ globally in the first round; qualified for the final round.

2024/02 **UK Integration Bee** - 6/42 in Cambridge (first round); 6/23 in the UK (second round).

2023/06 **STEP (Sixth Term Examination Paper)** - STEP 2: S (98/120); STEP 3: S (102/120).

Projects

2024/11-Present **CATAM Computational Mathematics Portfolio**

- Completed projects in numerical ODEs, finite fields, geometry, computational Galois theory, permutation groups, statistical inference, rigid body dynamics, and vortex ring simulation.
- Built Python/MATLAB pipelines for numerical integration, algebraic algorithms, simulation, and statistical estimation, assessing stability, convergence, complexity, and error.
- Processed and analysed numerical and statistical data in Python, visualising results and assessing error.

2020-Present **Desmos Mathematical Visualisation Projects** [GitHub]

- Created procedural visualisations with analytic functions, parametric equations, transformations, and projection geometry; built a 3D-style framework for depth cues, camera motion, and animation.

Experience

2026/08 **Peking University Summer School on Differential Geometry** - organised by Prof. Gang Tian

- Systematically studied Riemannian geometry, complex geometry and second-order elliptic PDEs, including Toponogov and volume comparison theorems, Kähler geometry and Hodge theory, Kodaira vanishing, and Schauder and De Giorgi-Nash estimates.
- Engaged in active discussions with peers and lecturers during special topics courses on interior regularity of minimal surfaces, Bernstein's theorem, Ricci flow singularities and solitons, and the role of Ricci flow in proving the Poincaré and geometrisation conjectures.

2023/07-Present **Private Tutor (100h+ high school maths, 40h+ topology)**

- Tutor A-Level, IB Mathematics and STEP; design personalised notes, problem sets and revision plans, and conduct Oxbridge-style mock interviews to develop clear reasoning under time pressure.
- Tutored two University of Leeds students in quotient spaces, connectedness and compactness; both passed their resit examinations.

Skills and Interests

Technical: Python, pandas, NumPy, Matplotlib, MATLAB, LaTeX, Data Structures, Advanced Algorithms.
Languages: Chinese (native), English (fluent, TOEFL iBT 110/120).
Go: Experience playing the strategic board game.
Frisbee: Team member.

References are available upon request.